QUANTUM
PORTFOLIO
CO-PILOT

WISER x Vanguard
Quantum for Finance
Challenge 2026

Quantum-Enhanced Multi-Asset Portfolio Construction

Anurag Sarkar | Ankit Gill

Audited classical optimization + exact QUBO/QAOA benchmark + higher-moment HUBO/VQE research extension

ZERO HARD-CONSTRAINT BREACHES **EXACT CLASSICAL VALIDATION** **QUANTUM-COMPATIBLE**

github.com/anurag-dot-physx/wiser-challenge-final

Claim boundary: quantum results are benchmarked against exact enumerable references; no quantum-advantage claim.

Challenge and scoring priorities

Optimize expected utility while preserving hard investment guardrails and interpretability.

Portfolio utility

Expected return + variance risk + income + implementation cost + scenario/drawdown control.

Hard guardrails

Fully invested, long-only, single-asset cap, equity maximum, defensive minimum, alternatives maximum.

Quantum compatibility

Binary variables, linear constraints and a quadratic cost function that can be audited classically.

What the solution must show

- Classical baseline, then constraints and scenario penalties
- Tunable growth, income, drawdown and cost sensitivity
- Risk/return/turnover/breach comparison with rationale
- Working co-pilot and classical validation

Design principle

Feasibility first. Every production recommendation is post-validated; every quantum-compatible encoding is compared with an exact classical reference.

Source: WISER <> Vanguard Quantum for Finance challenge brief.

Solution architecture

Separate the production recommendation from research benchmarks so comparisons remain valid.

Inputs

Synthetic/
anonymized
assumptions
+ investor controls

→

Production

8-asset convex QP
+ explicit guardrails

→

Audit

Continuous QP
+ exact 10% grid

→

Reduced quantum

Bounded bits + slack
+ exact QUBO +
QAOA

→

Research extension

Higher moments
+ CMA-ES + VQE
+ quadratization

Key separation

The eight-asset production solution is never compared directly with the reduced 5-sleeve quantum encoding. Each quantum result is judged against its own exact optimum.

Flagship model: convex quadratic utility + linear guardrails

$$J(w) = -\lambda\mu\ \mu^{\top}w + \lambda r\ w^{\top}\Sigma w - \lambda y\ y^{\top}w$$
$$+\ \lambda T\ (w-w_0)^{\top}R(w-w_0) + \lambda s \cdot \mathrm{mean}[(\ell s^{\top}w)^2]$$

Convexity audit

Σ, R and the scenario Gram matrix are positive semidefinite; the Hessian is positive semidefinite for every canonical profile.

Hard linear guardrails

- $\Sigma_i\ w_i = 1$ and $w_i \geq 0$
- single-asset maximum
- equity exposure maximum
- defensive exposure minimum
- alternatives exposure maximum

Investor controls

Growth · risk · income · drawdown · rebalancing sensitivity

Reported metrics

Expected return · volatility · risk-adjusted ratio · turnover · cost · scenario loss · breaches

Classical audit: exact enumeration verifies the production model

The 10% grid contains 19,448 fully invested portfolios.

Profile	Form error	min eig(H)	QP→grid gap	QUBO vars	Feasible
Growth	5.55e-17	1.010e-02	0.00081799	13	108
Balanced	6.94e-17	2.185e-02	0.00066947	15	66
Defensive	4.86e-17	3.365e-02	0.00140581	11	2

PASS

Zero canonical hard breaches. QUBO ground portfolio matches its exact reduced financial optimum.

Cost-sensitivity check

One-way turnover changes from 40% at $\lambda T = 0$ to 15% at $\lambda T = 2$, demonstrating a material implementation-cost control.

Interpretation

Objective decomposition matches matrix form to numerical precision; the continuous solution is a lower bound to the exact-grid objective, as expected.

Reduced exact-constraint QUBO and QAOA benchmark

$$E(x,s) = J(Ax) + P \sum_k (c_k^T x + d_k^T s_k - t_k)^2$$

Growth

13 vars
108 feasible

Balanced

15 vars
66 feasible

Defensive

11 vars
2 feasible

Exhaustive checks

- constraint-consistent states imply hard-feasible portfolios
- unique consistent portfolios = unique hard-feasible portfolios
- global QUBO ground is hard-feasible
- global QUBO ground matches reduced exact optimum

QAOA benchmark

Run only after the exact QUBO audit passes. Finite-shot post-processing selects among observed hard-feasible portfolios; fallback, probability mass and optimality gap are explicit.

Portfolio co-pilot: optimization translated into investor-facing decisions

Controls

Profile
Growth priority
Risk control
Income priority
Drawdown control
Cost sensitivity
Exposure guardrails

Allocation

Current vs continuous QP vs exact grid

Risk / return

Expected return, volatility, risk-adjusted ratio

Implementation

One-way turnover, gross turnover, estimated cost

Guardrails

PASS/FAIL checks for all hard constraints

Scenarios

Worst modeled scenario loss

Rationale

Largest allocations and changes in plain language

The co-pilot is the business-facing layer; exact audits remain underneath every recommendation.

Higher-moment extension: learn the Hamiltonian, then benchmark VQE

$$E(m) = -\lambda\mu\ \mu^{\mathsf{T}} m + \lambda\Sigma\ m^{\mathsf{T}}\Sigma m - \lambda S\ \Sigma S_{ijk} m_i m_j m_k + \lambda K\ \Sigma K_{ijkl} m_i m_j m_k m_l + \lambda B(\Sigma m_i - 8)^2$$

12 qubits

4 assets × 3 bits. Full source tensors. USD 1,250 unit makes 8 units exactly USD 10,000.

15 qubits

5 assets × 3 bits. Scaling demo; mixed TSLA+NVDA higher-moment terms are zero-imputed.

TRAIN

CMA-ES learns λ

→

SELECT

Choose G* on 5 validation sectors

→

HOLD OUT

Freeze model; test 5 untouched sectors

→

QUANTUM

Sweep final VQE budget

Efficiency: state features are cached once; VQE is intentionally outside the CMA-ES coefficient-training loop.

Exact HUBO → QUBO quadratization

Original HUBO

Cubic + quartic pseudo-Boolean interactions
12 or 15 native allocation bits

Rosenberg product constraint

Introduce $y = ab$
Add $P(ab - 2ay - 2by + 3y)$
Choose P above a rigorous objective range bound

Energy equivalence

Every original state has identical energy after consistent ancilla lifting.

Ground-state equivalence

Lifted QUBO ground portfolio is checked against the exact HUBO ground.

Scalability trade-off

Higher-order couplers are removed at the cost of additional binary variables.

Findings, limitations and reviewer path

Established

Convex flagship objective
Complete 19,448-state grid audit
Zero canonical hard breaches
Exact reduced-QUBO ground match

Exploratory

QAOA/VQE solver quality
Higher-moment calibration stability
15-qubit stitched scaling
Ancilla overhead

Claim boundary

No production quantum speedup claim
No economic outperformance claim
No restricted Vanguard data
Not investment advice

Reproduce the flagship audit

```
git clone https://github.com/anurag-dot-physx/wiser-challenge-final.git
pip install -r requirements.txt && pip install -e .
pytest -q && python src/audit_quadratic_model.py
streamlit run src/vanguard_copilot_app.py
```

Submission takeaway

Feasible first.
Exact reference second.
Quantum benchmark third.